

ME 4030: Final Project

Project Handout

Quarter-Car Active Suspension
Corrected public edition, September 2026

What to submit

Submit one PDF containing your derivations, controller design, MATLAB code, plots, performance tables, and discussion. Show intermediate steps and state assumptions. Do not submit a ZIP file. Task weights are 30, 30, and 40 points.

Problem statement. Design active suspension feedback for the quarter-car model below. The body (sprung mass) and wheel assembly (unsprung mass) move vertically about static equilibrium; gravity is already balanced. The actuator applies $+u$ to the body and $-u$ to the wheel. Use linear springs, viscous dampers, and an ideal force actuator, and discuss the limitations of these assumptions.

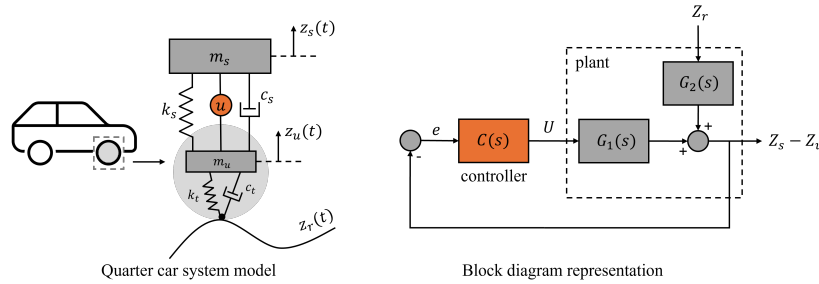


Figure 1: Original project system diagram. Displacements share one positive direction.

Parameter	Value	Meaning
M_s	2500 kg	Sprung mass
M_u	320 kg	Unsprung mass
K_s	80000 N/m	Suspension stiffness
C_s	350 N s/m	Suspension damping
K_t	500000 N/m	Tire stiffness
C_t	15020 N s/m	Tire damping

Define suspension deflection $x_s = z_s - z_u$ and tire deflection $x_h = z_u - z_r$. The equations of motion are

$$M_s \ddot{z}_s = -K_s(z_s - z_u) - C_s(\dot{z}_s - \dot{z}_u) + u, \quad (1)$$

$$M_u \ddot{z}_u = K_s(z_s - z_u) + C_s(\dot{z}_s - \dot{z}_u) - K_t(z_u - z_r) - C_t(\dot{z}_u - \dot{z}_r) - u. \quad (2)$$

Use zero initial conditions and a $T = 10$ s record. Evaluate both

$$z_r(t) = 0.20 \mathbf{1}_{t \geq 1} \text{ m}, \quad z_r(t) = 0.10 \sin(2\pi 0.5t) \text{ m}. \quad (3)$$

An ideal road step with tire damping produces a jump in wheel velocity. Resolve the fast transient rather than approximating the road derivative with a coarse difference.

Performance metrics and reporting

1. **Ride comfort:** absolute body-acceleration RMS,

$$a_{\text{RMS}} = \sqrt{\frac{1}{T} \int_0^T \ddot{z}_s(t)^2 dt}. \quad (4)$$

This is not acceleration relative to the road.

2. **Road holding:** tire-deflection RMS,

$$x_{h,\text{RMS}} = \sqrt{\frac{1}{T} \int_0^T [z_u(t) - z_r(t)]^2 dt}. \quad (5)$$

This linear-model proxy does not itself prove continuous tire contact.

3. **Transient performance:** for the step, report suspension recovery time: the first time after onset beyond which $|x_s| \leq 0.004$ m for the remainder of the record. Measure from $t = 1$ s. Report “not settled” if necessary. Percentage overshoot relative to $x_s(\infty) = 0$ is undefined.
4. **Suspension travel:** report $\max_{0 \leq t \leq T} |x_s(t)|$.
5. **Control effort:** report force RMS and peak force,

$$u_{\text{RMS}} = \sqrt{\frac{1}{T} \int_0^T u(t)^2 dt}, \quad u_{\text{peak}} = \max_{0 \leq t \leq T} |u(t)|. \quad (6)$$

Force RMS is not actuator energy; mechanical power also depends on relative velocity.

Metric	Excellent	Good	Acceptable
a_{RMS} (m/s ²)	≤ 0.35	≤ 0.50	≤ 0.70
$x_{h,\text{RMS}}$ (m)	≤ 0.008	≤ 0.012	≤ 0.020
Suspension recovery (s), step only	≤ 3	≤ 5	≤ 7
$\max x_s $ (m)	≤ 0.015	≤ 0.025	≤ 0.040
u_{RMS} (kN)	≤ 1.5	≤ 2.5	≤ 3.5

These are instructional design benchmarks. A single gain set need not meet all of them. Identify unmet targets and justify tradeoffs rather than claiming feasibility without evidence.

Supplemental body step metrics. Report 10–90% rise time, overshoot $100 \max(0, \max z_s - 0.20)/0.20$, and settling to 0.20 ± 0.004 m. Rise time is the difference between the first post-onset crossings of 0.02 and 0.18 m; report unavailable crossings explicitly. The original overshoot categories (2%, 5%, 10%) can be applied to *body motion*, not zero-final-value suspension deflection. Do not apply step rise, overshoot, or settling metrics to the sine run.

Comparison table. Make separate step and sine tables with columns for passive, PID, LQR, and achieved target category. Include all applicable metrics above, K_p, K_i, K_d, Q, R , and all five entries of the augmented LQR gain. State the sample interval, integration method, and a numerical convergence check. A 1 ms plot grid can miss significant energy in a fast PID transient; integrate squared outputs accurately.

Task 1 (30 pts): Design a PID controller

- (a) Derive $G_1(p) = [Z_s(p) - Z_u(p)]/U(p)$ with $Z_r = 0$. Use p for the Laplace variable and the output sign $x_s = z_s - z_u$ throughout.
- (b) Derive $G_2(p) = [Z_s(p) - Z_u(p)]/Z_r(p)$ with $U = 0$.
- (c) Simulate the passive system ($u = 0$) for both specified road inputs. Use MATLAB and show physical body/wheel displacement and velocity, suspension deflection, tire deflection, and road input.
- (d) Design negative-feedback PID control on x_s , with zero desired deflection. Report K_p, K_d, K_i , the feedback equation, and any derivative filter. MATLAB's `pidTuner` is optional; its reference-tracking step plot is not the road-disturbance response. Check closed-loop stability.
- (e) Simulate the PID response to the 0.20 m road step and compute the metrics. You may present several tuning sets if one set cannot meet all targets. Discuss the force spike caused by ideal derivative action and tire damping.
- (f) Simulate $z_r = 0.10 \sin(2\pi 0.5t)$ and discuss the PID performance using the applicable RMS, peak-travel, and force metrics.

Task 2 (30 pts): Design an LQR controller

- (a) Derive the physical-coordinate model with $q = [z_s, \dot{z}_s, z_u, \dot{z}_u]^T$:

$$\dot{q} = \bar{A}q + B_u u + B_{z_r} z_r + B_{\dot{z}_r} \dot{z}_r. \quad (7)$$

- (b) Construct $x = [z_s, \dot{z}_s, x_s, \eta]^T$ and obtain $\dot{x} = Ax + B_1 u + B_2 z_r$, $x_s = Cx$, eliminating the *road derivative*, not the road disturbance. Use

$$\alpha = \frac{C_s}{M_s} + \frac{C_s + C_t}{M_u}, \quad \eta = \dot{x}_s - \frac{C_t}{M_u} z_s + \alpha x_s + \frac{C_t}{M_u} z_r. \quad (8)$$

Explain why $\eta \neq \dot{x}_s$. Determine stability and controllability of (A, B_1) .

- (c) Simulate the open-loop step and sine responses using the transformed model. Recover physical wheel velocity correctly and compare with Task 1.
- (d) Augment with $\dot{\xi} = Cx$ and $x_a = [x^T, \xi]^T$. Choose $Q \succeq 0$, $R > 0$ and use `lqr` on the augmented plant. Report the five-entry gain, cost, closed-loop poles, and the meaning of your weights. Explain integral action.
- (e) Simulate the *LQR* response to the 0.20 m step and compute the metrics. Experiment with weights and explain both achieved and unmet targets.
- (f) Simulate the specified sinusoidal road and compare LQR with PID and passive responses.

Task 3 (40 pts): Performance analysis and discussion

- (a) Analyze the ride-comfort and road-holding tradeoffs achieved by PID and LQR. Use quantitative evidence from both disturbance cases.
- (b) Examine suspension travel and actuator force for both controllers. Discuss force, speed, stroke, bandwidth, saturation, derivative filtering, and integral windup in a practical implementation.
- (c) Which controller gives a more robust response to varying road disturbances, and why? Distinguish nominal performance from robustness. Support a robustness claim with additional tests (for example, road-frequency or mass/damping variations), or state clearly what the two nominal simulations cannot establish.
- (d) Could AI or data-driven methods adjust gains or predict the road ahead in an “autopilot” suspension? Discuss benefits, sensing/training data, validation, reliability, and safe fallback behavior.

Expected learning outcomes

By completing the project, you should be able to derive a mechanical plant and its transfer functions; interpret transformed states and disturbance channels; implement PID and integral-augmented LQR feedback; quantify performance and numerical error; and communicate the compromises and practical limits of a control design.

Working files

Start with `starter/main_file.m`. The reference MATLAB implementation and worked solutions are in `solutions/`; use MATLAB with Control System Toolbox. The separate solution document follows Tasks 1–3 and includes the executable MATLAB source in the original project’s colored, numbered listing style.

Corrections to the original offering

This edition preserves the original diagram, task sequence, six PID questions, six LQR questions, four discussion questions, and grading weights. It makes the deflection sign consistent, corrects the auxiliary state, separates body overshoot from suspension recovery, and corrects the LQR task’s mistaken reference to PID. The original Fall 2025 assignment dates are not deadlines for this public edition.